Pranav Bapu Hasban

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EDUCATION

Vellore Institute of Technology — Vellore, Tamil Nadu
B.Tech in Information Technology | Expected Graduation: 2028 | CGPA: 8.5

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL
Frontend: HTML, CSS, React.js, Redux Toolkit, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
Databases: MongoDB, SQL, DynamoDB
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, DynamoDB, Cognito, API Gateway, CloudWatch, SNS), Docker
DSA & Problem Solving: Data Structures & Algorithms, 250+ LeetCode problems (Rating 1450), Codeforces (Rating 1080+)
System Design: Cloud architecture on AWS, REST API design, database schema design, authentication and role-based access
Data Science / ML: Python, PyTorch, TensorFlow, MediaPipe, Pandas, NumPy, LSTM and CNN models
Tools & Deployment: Git, GitHub, Postman, VS Code, Render, Vercel, Cloudinary

PROJECTS

NexCart — Full-Stack E-Commerce Platform
React, Node.js, Express, MongoDB, Redux Toolkit, PayPal API, Cloudinary, JWT | GitHub | Live Demo

- Integrated a real-time AI chatbot sidebar powered by Gemini API with streaming responses, context-aware product recommendations, cart assistance, and order support, using a custom Express streaming endpoint and React ReadableStream
- Built a production-ready full-stack e-commerce platform with role-based access control and JWT authentication (custom middleware, guest-to-user cart merging, protected routes)
- Integrated PayPal Sandbox API for secure payment processing with full order lifecycle management
- Built an admin dashboard for real-time management of products, orders, users, and inventory; used Cloudinary for cloud image storage
- Independently deployed backend on Render and frontend on Vercel with environment-based configuration

AURA — Autonomous University Resource Allocator (Cloud Architecture Course Project, Team of 3)
React, Node.js, Express, AWS (EC2, S3, Lambda, DynamoDB, Cognito), TensorFlow | GitHub | Live Demo

- Built the React frontend for a green-cloud framework that forecasts university data-center demand and schedules servers and batch jobs by grid carbon intensity
- Developed an admin dashboard and interactive simulator where users set server count, power draw, flexible workload share and calendar events, compare saved scenarios, and export CSV and printable reports
- Deployed on AWS free tier with Cognito auth, an EC2-hosted Express API, DynamoDB, and CloudWatch/SNS alerts
- Simulated on real building-load data (CU-BEMS, 5.4M readings): the team's LSTM forecast had 20% lower error than a weekly baseline, and scheduling cut data-center energy by 39.5% and CO_2 by 41.3% versus always-on, with 99.5% SLA compliance

Multimodal Emotion Detection — Web App (Student Research Prototype) — In Progress
React, FastAPI, PyTorch, MediaPipe, WebSockets, SQLite

- Built a full-stack emotion recognition app that classifies 7 emotions from images, video clips, and live webcam streams (WebSocket) using ResNet-18 and CNN models trained on FER-2013
- Implemented a face pipeline (MediaPipe detection, square crop, model preprocessing) with 5 fps video sampling and moving-average smoothing; frames are processed in memory and never stored
- Designed a modality registry and confidence-weighted late fusion so audio, text, and EEG models can be plugged in without API changes; added a pytest suite and a session history view

CERTIFICATIONS

- OCI AI Foundations Associate — Oracle | 2025

ACHIEVEMENTS

- **Finalist** — International Innovation Challenge 3, Manipal University Jaipur (2026) | GitHub
- Built SPECTR, a drone defense prototype: an ESP32 drone node streams AES-encrypted sensor telemetry (GPS, tilt, obstacle distance, temperature) over MQTT to a Python backend with YOLOv8 weapon detection, which triggers email alerts to operators and feeds a live React dashboard